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INTERCONNECT STRUCTURES INCLUDING AIR GAPS

Abstract

A method and structure for forming a barrier-free interconnect layer includes patterning a metal layer disposed over a substrate to form a patterned metal layer including one or more trenches. In some embodiments, the method further includes selectively depositing a barrier layer on metal surfaces of the patterned metal layer within the one or more trenches. In some examples, and after selectively depositing the barrier layer, a dielectric layer is deposited within the one or more trenches. Thereafter, the selectively deposited barrier layer may be removed to form air gaps between the patterned metal layer and the dielectric layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/806,726, filed Jun. 13, 2022, issuing as U.S. Pat. No. 12,308,286, which is a divisional of U.S. patent application Ser. No. 16/788,057, filed Feb. 11, 2020, now U.S. Pat. No. 11,361,989, the entireties of which are incorporated by reference herein.

BACKGROUND

[0002] The electronics industry has experienced an ever-increasing demand for smaller and faster semiconductor devices which are simultaneously able to support a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). Thus far these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such scaling has also introduced increased complexity to the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

[0003] As merely one example, the formation of high-quality interconnects, including reliable metal lines and vias, has proved challenging. In particular, with the continued scaling of IC dimensions, and the corresponding reduction of layer thicknesses (e.g., metal, dielectric, and barrier layer thicknesses), parasitic resistance and capacitance (and the associated RC delay) of metal interconnect layers has become a critical issue. For example, in some existing metallization techniques, diffusion barrier layers (e.g., such as TaN) remaining after metallization can cause increased resistance. In addition, such diffusion barrier layers, which may have high dielectric constants, typically surround metal lines and cause increased capacitance. Further, after deposition of an inter-metal dielectric (IMD) layer, at least some existing techniques require a chemical mechanical polishing (CMP) process. CMP can be an expensive process, especially for IMD layers, which are generally harder than the metal used for metallization.

[0004] Thus, existing techniques have not proved entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present disclosure are best understood from the following detailed description when they are read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIG. 1 is a flow chart of a method of forming a barrier-free interconnect layer, in accordance with some embodiments;

[0007] FIGS. 2, 3, 4, 5, 6, 7, 8, 9, and 10 provide cross-sectional views of a device at intermediate stages of fabrication and processed in accordance with the method of FIG. 1;

[0008] FIG. 11 is a cross-sectional view of an MOS transistor according to some embodiments; and

[0009] FIG. 12 is perspective view of an embodiment of a FinFET device according to one or more

aspects of the present disclosure.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0011] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0012] It is also noted that the present disclosure presents embodiments in the form of barrier-free interconnect layers employed within a back-end-of-line (BEOL) process where a multi-level metal interconnect network is fabricated. In some embodiments, the barrier-free interconnect layers described herein may be employed within a local interconnect structure, an intermediate interconnect structure, and/or a global interconnect structure. As used herein, the term “local interconnect” is used to describe the lowest level of metal interconnects and are differentiated from intermediate and/or global interconnects. Local interconnects span relatively short distances and are sometimes used, for example, to electrically connect a source, drain, body, and/or gate of a given device, or those of nearby devices. Additionally, local interconnects may be used to facilitate a vertical connection of one or more devices to an overlying metallization layer (e.g., to an intermediate interconnect layer), for example, through one or more vias. Interconnects (e.g., including local, intermediate, or global interconnects), in general, may be formed as part of a BEOL fabrication processes and include a multi-level network of metal wiring. One of ordinary skill may recognize other embodiments of interconnect layers that may benefit from aspects of the present disclosure.

[0013] In addition, and in some embodiments, the techniques described herein and including the disclosed barrier-free interconnect layers may be employed within other semiconductor structures, circuits, and devices such as planar bulk metal-oxide-semiconductor field-effect transistors (MOSFETs), complementary MOS (CMOS) devices, multi-gate transistors (planar or vertical) such as FinFET devices, gate-all-around (GAA) devices, Omega-gate (Ω -gate) devices, or Pi-gate (Π -gate) devices, as well as strained-semiconductor devices, silicon-on-insulator (SOI) devices, partially-depleted SOI devices, fully-depleted SOI devices, memory devices such as flash memory (e.g., NAND or NOR flash memory), logic circuits, or other structures, circuits, or devices. One of ordinary skill may recognize other embodiments of semiconductor structures, circuits, or devices that may benefit from aspects of the present disclosure. Moreover, any of a plurality of IC circuits and/or devices may be connected by interconnects formed during a BEOL process.

[0014] With the aggressive scaling and ever-increasing complexity of advanced semiconductor devices and circuits, the formation of high-quality interconnects, including reliable metal lines and vias, has proved challenging. In particular, with the continued scaling of IC dimensions, and the corresponding reduction of layer thicknesses (e.g., metal, dielectric, and barrier layer thicknesses),

parasitic resistance and capacitance (and the associated RC delay) of metal interconnect layers has become a critical issue. For example, in some existing metallization techniques, diffusion barrier layers (e.g., such as TaN) remaining after metallization can cause increased resistance. In addition, such diffusion barrier layers, which may have high dielectric constants, typically surround metal lines and cause increased capacitance. Further, after deposition of an inter-metal dielectric (IMD) layer, at least some existing techniques require a chemical mechanical polishing (CMP) process. CMP can be an expensive process, especially for IMD layers, which are generally harder than the metal used for metallization. Thus, existing methods have not been entirely satisfactory in all respects.

[0015] Embodiments of the present disclosure offer advantages over the existing art, though it is understood that other embodiments may offer different advantages, not all advantages are necessarily discussed herein, and no particular advantage is required for all embodiments. For example, embodiments discussed herein include a barrier-free interconnect layer, and related methods, that effectively serve to overcome various shortcomings of existing methods. In at least some embodiments, a metal interconnect layer is patterned to form trenches in the metal interconnect layer within which an IMD layer will be deposited. Prior to IMD layer deposition, a selectively deposited barrier layer is formed on sidewalls, but not along a bottom surface, of the trenches in the metal interconnect layer and thereby reduces parasitic resistance and associated RC delay. In some examples, the selectively deposited barrier layer may include a self-assembled monolayer (SAM) or a polymer layer. In various embodiments, the SAM or polymer layer may be deposited using a vapor process (e.g., such as a chemical vapor deposition process, an atomic layer deposition process, or a molecular layer deposition process) or by a wet process (e.g., such as spin-on coating, tape casting, spraying, or dipping). In some embodiments, the SAM or polymer layer may include a specific functional group (e.g., selected based on the type of metal material of the metal interconnect layer), that causes the SAM or polymer layer to be selectively deposited onto metal surfaces of the patterned metal interconnect layer. In some cases, the functional group includes phosphonic acid (e.g., such as octadecylphosphonic acid), organosulfurs, hydroxide, or thiols (e.g., dodecanethiol, alkanethiol). As one example, if the metal material of the metal interconnect layer includes cobalt (Co), then phosphonic acid may be used as the functional group. If the metal material of the metal interconnect layer is copper (Cu), then a thiol may be used as the functional group.

[0016] Further, embodiments of the present disclosure include use of a catalyst layer for rapid atomic layer deposition (ALD) of the IMD layer. By employing the catalyst layer, a duration of the ALD IMD layer deposition process can be reduced by around 90%, thereby increasing processing throughput. In addition, the disclosed ALD IMD layer deposition process provides excellent thickness control and conformity of the IMD layer. As a result, and according to various embodiments, there is no need to perform a CMP process of the IMD layer, thus avoiding costly and difficult processing. After deposition of the IMD layer, and in some examples, the selectively deposited barrier layer (e.g., the SAM or polymer layer) can be removed to form an air gap between the patterned metal interconnect layer and the ALD-deposited IMD layer. Thus, rather than having a barrier layer with a high dielectric constant surrounding metal interconnect lines, embodiments of the present disclosure provide air gaps (e.g., with a dielectric constant of one (1)) surrounding the metal interconnect lines and thereby greatly reducing parasitic capacitance and associated RC delay. Further, and according to various embodiments, because the metal interconnect lines are not in contact with the IMD layer, but are instead separated by the air gaps, metal will not diffuse into the IMD layer. Therefore, an additional diffusion barrier layer is not needed. Additional details of embodiments of the present disclosure are provided below, and additional benefits and/or other advantages will become apparent to those skilled in the art having benefit of the present disclosure.

[0017] Referring now to FIG. 1, illustrated is a method **100** of forming a barrier-free interconnect

layer, in accordance with some embodiments. The method **100** is described below in more detail with reference to FIGS. **2-10**. It will be understood that additional process steps may be implemented before, during, and after the method **100**, and some process steps described may be replaced or eliminated in accordance with various embodiments of the method **100**. It will be further understood that parts of the method **100** may be fabricated by a well-known complementary metal-oxide-semiconductor (CMOS) technology process flow, and thus some processes are only briefly described herein.

[0018] The method **100** begins at block **102** where a substrate including one or more semiconductor devices is provided. With reference to FIG. **2**, and in an embodiment of block **102**, a device **200** including a substrate **202** is provided, where the substrate **202** includes one or more semiconductor devices. In some embodiments, the substrate **202** and the semiconductor devices therein may include devices such as those described below with reference to FIGS. **11** and **12**. By way of example, the semiconductor devices formed within the substrate **202** may be formed as part of a front-end-of-line (FEOL) process.

[0019] The method **100** proceeds to block **104** where a metal layer is formed as part of an interconnect network. Still referring to FIG. **2**, and in an embodiment of block **104**, a portion of a multi-level metal interconnect network, including a metal layer **204**, may be formed over the substrate **202** (e.g., as part of a BEOL process). In some cases, the metal layer **204** may include a portion of a metal line (of the multi-level metal interconnect network) that includes a copper (Cu) layer, an aluminum (Al) layer, an aluminum copper (AlCu) alloy layer, a ruthenium (Ru) layer, a cobalt (Co) layer, or other appropriate metal layer. In other examples, the metal layer **204** may include a portion of a metal via (of the multi-level metal interconnect network) that includes a Cu layer, an Al layer, an AlCu alloy layer, a Ru layer, a Co layer, a tungsten (W) layer, or other appropriate metal layer. In some examples, the metal layer **204** may be deposited by physical vapor deposition (PVD), chemical vapor deposition (CVD), electrochemical plating (ECP), electroless deposition (ELD), ALD, or a combination thereof. In some embodiments, the metal layer **204** may have a thickness 'T1' in a range of about 30-50 nm. In various embodiments, and prior to the deposition of the metal layer **204**, additional openings and metal layers (e.g., which may include additional metal lines or metal vias) may be formed beneath the metal layer **204** so as to provide electrical connections between underlying semiconductor devices (e.g., within the substrate **202**) and the subsequently deposited metal layer **204**.

[0020] After formation of the metal layer **204**, the method **100** proceeds to block **106** where the metal layer is patterned to form one or more trenches in the metal layer **204** within which an IMD layer will be subsequently formed. With reference to FIGS. **2** and **3**, and in an embodiment of block **106**, the metal layer **204** may be patterned using a combination of photolithography (e.g., including photoresist deposition, exposure, and development) and etching (e.g., using a wet or dry etching process) to form trenches **302** within the metal layer **204**. As shown, the trenches **302** include sidewall surfaces **302S** and a bottom surface **302B**. In some embodiments, the etching process includes a reactive ion etching (RIE) process. The RIE process may, in some examples, be performed using a chlorine-based chemical (e.g., such as CHCl₃). In the example of FIG. **3**, the trenches **302** are shown as having a trapezoidal shape. However, in some embodiments, the trenches **302** may have a rectangular shape. In some cases, a critical dimension (CD) of the trenches **302** may have a value in a range of about 10-12 nm.

[0021] After formation of the trenches **302**, the method **100** proceeds to block **108** where a barrier layer is selectively deposited. With reference to FIGS. **3** and **4**, and in an embodiment of block **108**, a barrier layer **402** is deposited over the substrate **202**. In particular, the barrier layer **402** is selectively deposited on metal surfaces of the patterned metal layer **204**, including on top surfaces of the patterned metal layer **204** and along sidewall surfaces **302S** of the trenches **302**, but not on the bottom surface **302B**. In various embodiments, the barrier layer **402** includes an organic spacer layer such as a SAM or a polymer layer. In some embodiments, the organic spacer layer may

include a specific functional group (e.g., selected based on a material used for the metal layer **204**), that causes the barrier layer **402** to be selectively deposited onto metal surfaces of the patterned metal layer **204**, while not being deposited on the bottom surface **302B** of the trenches **302**. In some examples, the functional group includes phosphonic acid (e.g., such as octadecylphosphonic acid), organosulfurs, hydroxide, or thiols (e.g., dodecanethiol, alkanethiol). As one example, if the patterned metal layer **204** includes cobalt (Co), then phosphonic acid may be used as the functional group. If the patterned metal layer **204** includes copper (Cu), then a thiol may be used as the functional group. In various embodiments, the barrier layer **402** may be deposited using a vapor process (e.g., such as a chemical vapor deposition process, an atomic layer deposition process, or a molecular layer deposition process) or by a wet process (e.g., such as spin-on coating, tape casting, spraying, or dipping). In some embodiments, the barrier layer **402** may have a thickness 'T2' in a range of about 10-50Å. In some cases, the thickness of the barrier layer **402** may be substantially the same along both top surfaces of the patterned metal layer **204** and along sidewall surfaces **302S** of the trenches **302**. In an example, the barrier layer **402** may be deposited using a spin-coating process with a spin speed of about 400-1000 RPM, a spin duration of about 10 s, and a solution concentration of about 1 mM.

[0022] After selective deposition of the barrier layer **402**, the method **100** proceeds to block **110** where a catalyst layer is formed. With reference to FIGS. **4** and **5**, and in an embodiment of block **110**, a catalyst layer **502** is formed along the bottom surface **302B** of the trenches **302**. In some embodiments, the catalyst layer **502** includes a layer of trimethylaluminum (TMA). In some embodiments, the catalyst layer **502** is formed by exposing the device **200** to a precursor (e.g., such as TMA) for a soaking period. In some examples, a CVD or ALD system may be used to expose the device **200** to the precursor for the soaking period (e.g., as part of a CVD or ALD process). In some embodiments, the catalyst layer **502** has a thickness 'T3' of about one (1) molecular layer or less than about 5Å. The catalyst layer **502**, as shown in the example of FIG. **5**, is formed by adsorption of precursor molecules only on areas of the device **200** that are not covered by the barrier layer **402** (e.g., the bottom surface **302B** of the trenches **302**). In various embodiments, the catalyst layer **502** serves to catalyze the IMD layer deposition process, as described below.

[0023] The method **100** proceeds to block **112** where an IMD layer is deposited over the catalyst layer. With reference to FIGS. **5** and **6**, and in an embodiment of block **112**, an IMD layer **602** is deposited within the trenches **302** and over the catalyst layer **502**. In some embodiments, the IMD layer **602** is deposited by ALD. However, because of the catalyst layer **502**, the ALD deposition of the IMD layer **602** proceeds significantly more rapidly than would be possible without the catalyst layer **502**. In some examples, by employing the catalyst layer **502**, a duration of the ALD IMD layer deposition process may be reduced by around 90% as compared to ALD IMD layer deposition without the catalyst layer **502**. As such, processing throughput is markedly increased. In some embodiments, the IMD layer **602** may include SiO_x, SiCOH, boron carbide, or other appropriate material. By way of example, the IMD layer **602** may be formed by exposing the device **200** to an appropriate precursor (e.g., such as a SiO_x, SiCOH, or boron carbide precursor(s)) for a soaking period. In some cases, a top portion of the IMD layer **602** may include a small hump having a height 'H1' in a range of about 0-10Å. Regardless of the possibility of forming such a hump, the rapid ALD IMD layer deposition process provides excellent thickness control and conformity of the IMD layer **602**. As a result, and according to various embodiments, there is no need to perform a CMP process of the as-deposited IMD layer **602**, thus avoiding costly and difficult processing. Stated another way, the rapid ALD IMD layer deposition process used to form the IMD layer **602** provides for a substantially planar top surface of the device **200** after deposition of the IMD layer **602**.

[0024] In some embodiments, the IMD layer **602** may alternatively include a low-K dielectric layer (e.g., having a dielectric constant less than that of SiO₂ which is about 3.9) such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as

borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable low-K dielectric material. In some alternative techniques, the IMD layer **602** may be deposited by a subatmospheric CVD (SACVD) process, a flowable CVD process, or other suitable deposition technique.

[0025] After depositing the IMD layer **602**, the method **100** proceeds to block **114** where the barrier layer is removed to form air gaps between the patterned metal layer and the IMD layer. With reference to FIGS. **6** and **7**, and in an embodiment of block **114**, the selectively deposited barrier layer **402** is removed to form air gaps **702** between the patterned metal layer **204** and the IMD layer **602**. In some cases, the barrier layer **402** may be removed using an NH.sub.3 plasma treatment, an H.sub.2 plasma treatment, or other appropriate wet or dry etching process. By way of example, the NH.sub.3 or H.sub.2 plasma treatment may be performed at a power of between about 500-1200 W and for a duration of between about 10-15 s. In some embodiments, the plasma treatment may planarize a top surface of the IMD layer **602**. In contrast to some existing embodiments that include a high dielectric constant barrier layer surrounding metal interconnect lines, embodiments of the present disclosure provide the air gaps **702** (e.g., with a dielectric constant of one (1)) surrounding the patterned metal layer **204**. As a result, the parasitic capacitance and associated RC delay of the device **200** is greatly reduced. Further, and because the patterned metal layer **204** is not in contact with the IMD layer **602**, but is instead separated by the air gaps **702**, metal will not diffuse into the IMD layer **602**. Therefore, an additional diffusion barrier layer is not needed.

[0026] After formation of the air gaps **702**, the method **100** proceeds to block **116** where an etch stop layer (ESL) and an inter-layer dielectric (ILD) layer are deposited. With reference to FIGS. **7** and **8**, and in an embodiment of block **116**, an ESL **802** is deposited over the device **200**. The ESL **802** may comprise a single layer or multiple layers. In addition to providing an etch stop, the ESL **802** may also improve etching uniformity. In some embodiments, the ESL **802** may include one or more of AlOx, AlZrOx, ZrOx, SiCN, SiO, SiOC, or other appropriate material. In various examples, the ESL **802** may have a thickness in a range of about 30-100Å. In some cases, the ESL **802** may be deposited by ALD, CVD, PVD, or other appropriate deposition method.

[0027] Still referring to FIG. **8**, and in a further embodiment of block **116**, an ILD layer **804** is deposited over the ESL **802**. In some embodiments, the ILD layer **804** may include a dielectric material such as SiCOH, SiOx, or other appropriate material. In some embodiments, ILD layer **804** may alternatively include a low-K dielectric layer such as TEOS oxide, un-doped silicate glass, or doped silicon oxide such as BPSG, FSG, PSG, BSG, and/or other suitable low-K dielectric material. In various examples, the ILD layer **804** may have a thickness in a range of about 300-500Å. In some cases, ILD layer **804** may be deposited by ALD, CVD, PVD, SACVD, flowable CVD, or other suitable deposition technique.

[0028] After deposition of the ILD layer **804**, the method **100** proceeds to block **118** where the ESL **802** and the ILD layer **804** are patterned to form a via opening. With reference to FIGS. **8** and **9**, and in an embodiment of block **118**, the ESL **802** and the ILD layer **804** may be patterned using a combination of photolithography (e.g., including photoresist deposition, exposure, and development) and etching (e.g., using a wet or dry etching process) to form a via opening **902** within the ESL **802** and the ILD layer **804**. In the example of FIG. **9**, the via opening **902** is shown as having a trapezoidal shape. However, in some embodiments, the via opening **902** may have a rectangular shape.

[0029] The method **100** proceeds to block **120** where a diffusion barrier is deposited. Still with reference to FIG. **9**, and in an embodiment of block **120**, a diffusion barrier **904** is deposited over the device **200** and within the via opening **902**, including along sidewalls and the bottom surface of the via opening **902**. In some embodiments, the diffusion barrier **904** is conformally deposited, thus a thickness of the diffusion barrier **904** over the device **200** is substantially uniform. In some examples, the diffusion barrier **904** may have a thickness in a range of about 10-30Å. The diffusion barrier **904** may be deposited by CVD, ALD, or PVD. In some cases, the diffusion barrier **904** may

include TaN, SiN, or other appropriate material.

[0030] After deposition of the diffusion barrier **904**, the method **100** proceeds to block **122** where a metal layer is deposited. With reference to FIGS. **9** and **10**, and in an embodiment of block **122**, a metal layer **1002** is deposited over the diffusion barrier **904** and within the via opening **902**. Thus, the metal layer **1002** provides electrical contact to the underlying metal layer **204**. In some embodiments, the metal layer **1002** may have a thickness in a range of about 30-50Å. In some cases, the metal layer **1002** may be deposited by ECP, ELD, PVD, ALD, or other appropriate process.

[0031] As discussed above, the semiconductor devices formed within the substrate **202** may include devices such as those described with reference to FIGS. **11** and **12**. In addition, while the above discussion presented embodiments for forming a barrier-free interconnect layer within a BEOL process, the techniques described herein may be employed to form barrier-free metal layers for metal layers formed as part of the fabrication of devices in a FEOL process, such as those described with reference to FIGS. **11** and **12**. As such, the devices of FIGS. **11** and **12** will now be discussed.

[0032] Referring first to the example of FIG. **11**, illustrated therein is an MOS transistor **1100**, providing an example of merely one device type which may include embodiments of the present disclosure. The transistor **1100** is fabricated on a substrate **1102** and includes a gate stack **1104**. The substrate **1102** may be a semiconductor substrate such as a silicon substrate. The substrate **1102** may include various layers, including conductive or insulating layers formed on the substrate **1102**. The substrate **1102** may include various doping configurations depending on design requirements as is known in the art. The substrate **1102** may also include other semiconductors such as germanium, silicon carbide (SiC), silicon germanium (SiGe), or diamond. Alternatively, the substrate **1102** may include a compound semiconductor and/or an alloy semiconductor. Further, in some embodiments, the substrate **1102** may include an epitaxial layer (epi-layer), the substrate **1102** may be strained for performance enhancement, the substrate **1102** may include a silicon-on-insulator (SOI) structure, and/or the substrate **1102** may have other suitable enhancement features.

[0033] The gate stack **1104** includes a gate dielectric **1106** and a gate electrode **1108** disposed on the gate dielectric **1106**. In some embodiments, the gate dielectric **1106** may include an interfacial layer such as silicon oxide layer (SiO₂) or silicon oxynitride (SiON), where such interfacial layer may be formed by chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable method. In some examples, the gate dielectric **1106** includes a high-k dielectric layer such as hafnium oxide (HfO₂). Alternatively, the high-k dielectric layer may include other high-k dielectrics, such as TiO₂, HfZrO₂, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, LaO, AlO, ZrO, TiO, Ta₂O₅, Y₂O₃, SrTiO₃ (STO), BaTiO₃ (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO₃ (BST), Al₂O₃, Si₃N₄, oxynitrides (SiON), combinations thereof, or other suitable material. High-K gate dielectrics, as used and described herein, include dielectric materials having a high dielectric constant, for example, greater than that of thermal silicon oxide (~3.9). In still other embodiments, the gate dielectric **1106** may include silicon dioxide or other suitable dielectric. The gate dielectric **1106** may be formed by ALD, PVD, CVD, oxidation, and/or other suitable methods.

[0034] In some embodiments, the gate electrode **1108** may be deposited as part of a gate first or gate last (e.g., replacement gate) process. In various embodiments, the gate electrode **1108** includes a conductive layer such as W, Ti, TiN, TiAl, TiAlN, Ta, TaN, WN, Re, Ir, Ru, Mo, Al, Cu, Co, CoSi, Ni, NiSi, combinations thereof, and/or other suitable compositions. In some examples, the gate electrode **1108** may include a first metal material for an N-type transistor and a second metal material for a P-type transistor. Thus, the transistor **1100** may include a dual work-function metal gate configuration. For example, the first metal material (e.g., for N-type devices) may include metals having a work function substantially aligned with a work function of the substrate conduction band, or at least substantially aligned with a work function of the conduction band of a

channel region **1114** of the transistor **1100**. Similarly, the second metal material (e.g., for P-type devices) may include metals having a work function substantially aligned with a work function of the substrate valence band, or at least substantially aligned with a work function of the valence band of the channel region **1114** of the transistor **1100**. Thus, the gate stack **1104** may provide a gate electrode for the transistor **1100**, including both N-type and P-type devices. In some embodiments, the gate electrode **1108** may alternately or additionally include a polysilicon layer. In various examples, the gate electrode **1108** may be formed using PVD, CVD, electron beam (e-beam) evaporation, and/or other suitable process. In some embodiments, sidewall spacers are formed on sidewalls of the gate stack **1104**. Such sidewall spacers may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, or combinations thereof.

[0035] The transistor **1100** further includes a source region **1110** and a drain region **1112** each formed within the semiconductor substrate **1102**, adjacent to and on either side of the gate stack **1104**. In some embodiments, the source and drain regions **1110**, **1112** include diffused source/drain regions, ion implanted source/drain regions, epitaxially grown regions, or a combination thereof. The channel region **1114** of the transistor **1100** is defined as the region between the source and drain regions **1110**, **1112** under the gate dielectric **1106**, and within the semiconductor substrate **1102**. The channel region **1114** has an associated channel length “L” and an associated channel width “W”. When a bias voltage greater than a threshold voltage ($V_{sub.t}$) (i.e., turn-on voltage) for the transistor **1100** is applied to the gate electrode **1108** along with a concurrently applied bias voltage between the source and drain regions **1110**, **1112**, an electric current (e.g., a transistor drive current) flows between the source and drain regions **1110**, **1112** through the channel region **1114**. The amount of drive current developed for a given bias voltage (e.g., applied to the gate electrode **1108** or between the source and drain regions **1110**, **1112**) is a function of, among others, the mobility of the material used to form the channel region **1114**. In some examples, the channel region **1114** includes silicon (Si) and/or a high-mobility material such as germanium, which may be epitaxially grown, as well as any of the plurality of compound semiconductors or alloy semiconductors as known in the art. High-mobility materials include those materials with electron and/or hole mobility greater than silicon (Si), which has an intrinsic electron mobility at room temperature (300 K) of around $1350 \text{ cm}^2/\text{V}\cdot\text{s}$ and an intrinsic hole mobility at room temperature (300 K) of around $480 \text{ cm}^2/\text{V}\cdot\text{s}$.

[0036] Referring now to FIG. 12, illustrated therein is a FinFET device **1200**, providing an example of an alternative device type which may include embodiments of the present disclosure. By way of example, the FinFET device **1200** includes one or more fin-based, multi-gate field-effect transistors (FETs). The FinFET device **1200** includes a substrate **1252**, at least one fin element **1254** extending from the substrate **1252**, isolation regions **1256**, and a gate structure **1258** disposed on and around the fin-element **1254**. The substrate **1252** may be a semiconductor substrate such as a silicon substrate. In various embodiments, the substrate **1252** may be substantially the same as the substrate **1102** and may include one or more of the materials used for the substrate **1102**, as described above.

[0037] The fin-element **1254**, like the substrate **1252**, may include one or more epitaxially-grown layers, and may comprise silicon or another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, and/or GaInAsP; or combinations thereof. The fins **1254** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (resist) overlying the substrate (e.g., on a silicon layer), exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a masking element including the resist. In some embodiments, patterning the resist to form the masking element may be performed using an electron beam (e-beam) lithography process. The masking element may then be used to protect regions of the

substrate while an etch process forms recesses into the silicon layer, thereby leaving an extending fin **1254**. The recesses may be etched using a dry etch (e.g., chemical oxide removal), a wet etch, and/or other suitable processes. Numerous other embodiments of methods to form the fins **1254** on the substrate **1252** may also be used.

[0038] Each of the plurality of fins **1254** also include a source region **1255** and a drain region **1257** where the source/drain regions **1255**, **1257** are formed in, on, and/or surrounding the fin **1254**. The source/drain regions **1255**, **1257** may be epitaxially grown over the fins **1254**. In addition, a channel region of a transistor is disposed within the fin **1254**, underlying the gate structure **1258**, along a plane substantially parallel to a plane defined by section AA' of FIG. **12**. In some examples, the channel region of the fin includes a high-mobility material, as described above.

[0039] The isolation regions **1256** may be shallow trench isolation (STI) features. Alternatively, a field oxide, a LOCOS feature, and/or other suitable isolation features may be implemented on and/or within the substrate **1252**. The isolation regions **1256** may be composed of silicon oxide, silicon nitride, silicon oxynitride, FSG, a low-k dielectric, combinations thereof, and/or other suitable material known in the art. In an embodiment, the isolation regions **1256** are STI features and are formed by etching trenches in the substrate **1252**. The trenches may then be filled with isolating material, followed by a CMP process. However, other embodiments are possible. In some embodiments, the isolation regions **1256** may include a multi-layer structure, for example, having one or more liner layers.

[0040] The gate structure **1258** includes a gate stack having an interfacial layer **1260** formed over the channel region of the fin **1254**, a gate dielectric layer **1262** formed over the interfacial layer **1260**, and a metal layer **1264** formed over the gate dielectric layer **1262**. In various embodiments, the interfacial layer **1260** is substantially the same as the interfacial layer described as part of the gate dielectric **1106**. In some embodiments, the gate dielectric layer **1262** is substantially the same as the gate dielectric **1106** and may include high-k dielectrics similar to that used for the gate dielectric **1106**. Similarly, in various embodiments, the metal layer **1264** is substantially the same as the gate electrode **1108**, described above. In some embodiments, sidewall spacers are formed on sidewalls of the gate structure **1258**. The sidewall spacers may include a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, or combinations thereof.

[0041] As discussed above, each of the transistor **1100** and FinFET device **1200** may include barrier-free metal layers formed as described above, where a selectively deposited barrier layer is deposited before a dielectric layer deposition adjacent to a metal layer, and where the selectively deposited barrier layer is subsequently removed to form air gaps between the dielectric layer and the adjacent metal layer.

[0042] The various embodiments described herein thus offer several advantages over the existing art. It will be understood that not all advantages have been necessarily discussed herein, no particular advantage is required for all embodiments, and other embodiments may offer different advantages. As one example, embodiments discussed herein include a barrier-free interconnect layer, and related methods, that effectively serve to overcome various shortcomings of existing methods. In at least some embodiments, and prior to IMD layer deposition, a selectively deposited barrier layer is formed on sidewalls, but not along a bottom surface, of trenches formed within a metal interconnect layer. In some examples, the selectively deposited barrier layer may include a SAM or a polymer layer. In various embodiments, the SAM or polymer layer may be deposited using a vapor processor by a wet process. In some embodiments, the SAM or polymer layer may include a specific functional group that causes the SAM or polymer layer to be selectively deposited onto metal surfaces of the patterned metal interconnect layer. In some cases, the functional group includes phosphonic acid, organosulfurs, hydroxide, or thiols.

[0043] Further, embodiments of the present disclosure include use of a catalyst layer for rapid ALD deposition of the IMD layer. By employing the catalyst layer, a duration of the ALD IMD layer deposition process can be reduced by around 90%, thereby increasing processing throughput. In

addition, the disclosed ALD IMD layer deposition process provides excellent thickness control and conformity of the IMD layer. As a result, there is no need to perform a CMP process of the IMD layer, thus avoiding costly and difficult processing. After deposition of the IMD layer, and in some examples, the selectively deposited barrier layer can be removed to form an air gap between the patterned metal interconnect layer and the ALD-deposited IMD layer. Thus, rather than having a barrier layer with a high dielectric constant surrounding metal interconnect lines, embodiments of the present disclosure provide air gaps surrounding the metal interconnect lines and thereby greatly reducing parasitic capacitance and associated RC delay. Further, and according to various embodiments, because the metal interconnect lines are not in contact with the IMD layer, but are instead separated by the air gaps, metal will not diffuse into the IMD layer. Therefore, an additional diffusion barrier layer is not needed. Additional benefits and/or other advantages will become apparent to those skilled in the art having benefit of the present disclosure.

[0044] Thus, one of the embodiments of the present disclosure described a method for fabricating a semiconductor device including patterning a metal layer disposed over a substrate to form a patterned metal layer including one or more trenches. In some embodiments, the method further includes selectively depositing a barrier layer on metal surfaces of the patterned metal layer within the one or more trenches. In some examples, and after selectively depositing the barrier layer, a dielectric layer is deposited within the one or more trenches. Thereafter, the selectively deposited barrier layer may be removed to form air gaps between the patterned metal layer and the dielectric layer.

[0045] In another of the embodiments, discussed is a method of fabricating a semiconductor device including forming a portion of a multi-level metal interconnect network including a first metal region and a second metal region. The first metal region and the second metal region are separated by a trench, where a first side of the first metal region defines a first trench sidewall, and where a second side of the second metal region defines a second trench sidewall opposing the first trench sidewall. In some embodiments, the method further includes depositing an organic spacer layer on the first trench sidewall and the second trench sidewall, while a trench bottom surface that interposes the first and second trench sidewalls remains substantially free of the organic spacer layer. By way of example, and after depositing the organic spacer layer, an IMD layer is formed within the trench and over the trench bottom surface. In some embodiments, and after forming the IMD layer, the organic spacer layer is etched to form a first air gap separating the first trench sidewall from the IMD layer and a second air gap separating the second trench sidewall from the IMD layer.

[0046] In yet another of the embodiments, discussed is a device including a substrate having one or more semiconductor devices. In some embodiments, the device further includes a metal interconnect layer disposed over the substrate and having a plurality of metal regions interposed by a plurality of dielectric layer regions. In various examples, each of the plurality of metal regions is separated from an adjacent dielectric layer region of the plurality of dielectric layer regions by an air gap. In some embodiments, the metal interconnect layer provides an electrical connection to the one or more semiconductor devices. The device may further include a catalyst layer interposing the substrate and each of the plurality of dielectric layer regions.

[0047] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

- 1.** A method of fabricating a semiconductor device, comprising: selectively depositing a self-assembled monolayer (SAM) on a patterned metal layer of a multi-level interconnect network; while the SAM remains on top metal surfaces of the patterned metal layer, depositing a dielectric layer in trenches distributed throughout the patterned metal layer; and performing a plasma treatment to remove the SAM and form air gaps between the patterned metal layer and the dielectric layer, wherein the plasma treatment planarizes a top surface of the dielectric layer.
- 2.** The method of claim 1, wherein the patterned metal layer is part of a lowest level of the multi-level interconnect network.
- 3.** The method of claim 1, wherein the patterned metal layer includes a metal line or a metal via of the multi-level interconnect network.
- 4.** The method of claim 1, wherein the trenches include sidewall surfaces and a bottom surface, and wherein the SAM is selectively deposited on the sidewall surfaces without being deposited on the bottom surface.
- 5.** The method of claim 1, wherein the SAM includes a functional group that causes the SAM to be selectively deposited on metal surfaces of the patterned metal layer.
- 6.** The method of claim 5, wherein the functional group includes phosphonic acid, organosulfurs, hydroxide, or thiols.
- 7.** The method of claim 4, further comprising: prior to depositing the dielectric layer, forming a catalyst layer on the bottom surface of the trenches; and after the forming the catalyst layer, depositing the dielectric layer on the catalyst layer within the trenches.
- 8.** The method of claim 7, wherein the depositing the dielectric layer includes depositing the dielectric layer using an atomic layer deposition (ALD) process, and wherein the catalyst layer serves to catalyze the ALD process.
- 9.** The method of claim 1, wherein prior to performing the plasma treatment to remove the SAM, a top portion of the dielectric layer within the trenches includes a hump that extends above the top metal surfaces of the patterned metal layer.
- 10.** A method of fabricating a semiconductor device, comprising: forming a first metal region and a second metal region separated by a trench, wherein a first width of a top of the trench is greater than a second width of a bottom of the trench, and wherein opposing sidewalls of the trench are defined by respective opposing sides of the first metal region and the second metal region; depositing a self-assembled monolayer (SAM) on the opposing sidewalls of the trench and on top surfaces of the first metal region and the second metal region; while the SAM remains on the top surfaces of the first metal region and the second metal region, forming an inter-metal dielectric (IMD) layer within the trench; and after the forming the IMD layer, removing the SAM to form respective first and second air gaps separating the opposing sidewalls of the trench from the IMD layer.
- 11.** The method of claim 10, wherein a bottom surface of the trench remains free of the SAM after the depositing the SAM.
- 12.** The method of claim 10, further comprising: prior to forming the IMD layer, forming a catalyst layer on a bottom surface of the trench; and after the forming the catalyst layer, forming the IMD layer within the trench and over the catalyst layer.
- 13.** The method of claim 12, wherein the catalyst layer includes a trimethylaluminium (TMA) layer.
- 14.** The method of claim 10, wherein the SAM includes a functional group that causes the SAM to be selectively deposited on the opposing sidewalls of the trench and on top surfaces of the first metal region and the second metal region.
- 15.** The method of claim 14, wherein the functional group includes phosphonic acid, organosulfurs,

hydroxide, or thiols.

16. The method of claim 10, wherein the removing the SAM includes removing the SAM using an NH.sub.3 plasma treatment or an H.sub.2 plasma treatment.

17. The method of claim 10, wherein the removing the SAM planarizes a top surface of the IMD layer.

18. A method, comprising: forming a barrier layer on sidewall surfaces and on top surfaces of a patterned metal interconnect, wherein bottom surfaces of trenches disposed in the patterned metal interconnect define regions free of the barrier layer; depositing a catalyst layer on the regions free of the barrier layer; while the barrier layer remains on the top surfaces of the patterned metal interconnect, depositing a dielectric layer within the trenches and over the catalyst layer, wherein a top portion of the dielectric layer extends above the top surfaces of the patterned metal interconnect; and after depositing the dielectric layer, removing the barrier layer to form air gaps between the dielectric layer and the patterned metal interconnect, wherein the removing the barrier layer removes the top portion of the dielectric layer.

19. The method of claim 18, wherein the catalyst layer includes a layer of trimethylaluminum (TMA).

20. The method of claim 18, wherein the barrier layer includes a self-assembled monolayer (SAM) or a polymer layer.
